

Лабораторная работа №7

Настройка Firewalld, перенаправления портов и Masquerading

Владимир Базлов

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Российский университет дружбы народов, Москва, Россия

Цель работы

Получить практические навыки настройки **Firewalld**,
создания пользовательских служб, настройки **Port Forwarding**
и включения **Masquerading** в Linux.

Работа с Firewalld

```
[root@server.vabazlov.net server]# cp /usr/lib/firewalld/services/ssh.xml /etc/firewalld/services/ssh-custom.xml
[root@server.vabazlov.net server]# cd /etc/firewalld/services/
[root@server.vabazlov.net services]# cat ssh-custom.xml
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>SSH</short>
  <description>Secure Shell (SSH) is a protocol for logging into and executing commands on remote machines. It provides secure encrypted communications. If you plan on accessing your machine remotely via SSH over a firewalled interface, enable this option. You need the openssh-server package installed for this option to be useful.</description>
  <port protocol="tcp" port="22"/>
</service>
[root@server.vabazlov.net services]# █
```

Рис. 1: Просмотр файла ssh-custom

Проверка и активация службы

```
[root@server.vabazlov.net services]# firewall-cmd --get-services
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client amqp amqps anno-1602 anno-1800 apcupsd aseqnet audit ausweisap
p2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent
t-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent civilization-iv civilization-v cockpit collectd condor-collector cratedb ctdb dds dds-mu
lticast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-qtcp dns-over-tls docker-registry docker-swarm dropbox-lansync elasticsearch et
cd-client etcd-server factorio finger foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera gan
glia-client ganglia-master git gpsd grafana gre high-availability http http3 https ident imap imaps iperf2 iperf3 ipfs ipp ipp-client ipsec irc i
rcs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogind kpasswd kprotop kshell kube-api kube-apiserver kube-control-plane kube-contr
ol-plane-secure kube-controller-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker ku
belet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmn llmn-client llmn-tcp llmn-udp managiesieve matrix
mdns memcache minecraft minidlna mndp mongodb mosh mountd mpd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula need-for-speed-most-wanted netbi
os-ns netdata-dashboard nfs nfs3 nmap-0183 nripe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storageconsole ovirt-vaconsole plex pncd pmpox
y pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster q
uassel radius radsec rdp redis redis-sentinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane settlers-histor
y-collection sip sips slimevr slp smtp smtp-submission snmpsnmp-tls snmp-tls-trap snmptrap spideroak-lansync spotify-sync squid ssdp ssh ssh
-custom statsrv steam-lan-transfer steam-streaming stellaris stronghold-crusader stun stuns submission supertuxkart svdrp svn syncthing syncthing
-gui syncthing-relay synergy sysconlan syslog syslog-tls telnet tentacle terraria tftp tile38 tinc tor-socks transmission-client turn turns upnp-
client vdsd vnc-server vrrp warpinator wben-http wben-https wireguard ws-discovery ws-discovery-client ws-discovery-host ws-discovery-tcp ws-disc
overy-udp wsdd wsdd-http wsmann wsmans xdncp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-java-gateway zabbix-server zabbix-tr
apper zabbix-web-service zero-k zerotier
[root@server.vabazlov.net services]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https ssh
[root@server.vabazlov.net services]# firewall-cmd --add-service=ssh-custom
success
[root@server.vabazlov.net services]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https ssh ssh-custom
[root@server.vabazlov.net services]# firewall-cmd --add-service=ssh-custom --permanent
success
[root@server.vabazlov.net services]# firewall-cmd --reload
success
[root@server.vabazlov.net services]#
```

Рис. 2: Список служб

```
[vabazlov@client.vabazlov.net ~]$ ssh -p 2022 vabazlov@server.vabazlov.net
The authenticity of host '[server.vabazlov.net]:2022 ([192.168.1.1]:2022)' can't
be established.
ED25519 key fingerprint is SHA256:n0w0vphoobfpKiXoFhmCpcTAKAVt01RYFK5Kxnw0XTU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[server.vabazlov.net]:2022' (ED25519) to the list of
known hosts.
vabazlov@server.vabazlov.net's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Mon Nov 24 06:59:47 2025
[vabazlov@server.vabazlov.net ~]$
[vabazlov@server.vabazlov.net ~]$
logout
Connection to server.vabazlov.net closed.
[vabazlov@client.vabazlov.net ~]$
```

Рис. 3: SSH на порт 2022

Включение forwarding

```
[root@server.vabazlov.net services]# sysctl -a | grep forward
net.ipv4.conf.all.bc_forwarding = 0
net.ipv4.conf.all.forwarding = 0
net.ipv4.conf.all.mc_forwarding = 0
net.ipv4.conf.default.bc_forwarding = 0
net.ipv4.conf.default.forwarding = 0
net.ipv4.conf.default.mc_forwarding = 0
net.ipv4.conf.eth0.bc_forwarding = 0
net.ipv4.conf.eth0.forwarding = 0
net.ipv4.conf.eth0.mc_forwarding = 0
net.ipv4.conf.eth1.bc_forwarding = 0
net.ipv4.conf.eth1.forwarding = 0
net.ipv4.conf.eth1.mc_forwarding = 0
net.ipv4.conf.lo.bc_forwarding = 0
net.ipv4.conf.lo.forwarding = 0
net.ipv4.conf.lo.mc_forwarding = 0
net.ipv4.ip_forward = 0
net.ipv4.ip_forward_update_priority = 1
net.ipv4.ip_forward_use_pmtu = 0
net.ipv6.conf.all.mc_forwarding = 0
net.ipv6.conf.all.forwarding = 0
net.ipv6.conf.default.mc_forwarding = 0
net.ipv6.conf.default.forwarding = 0
net.ipv6.conf.eth0.forwarding = 0
net.ipv6.conf.eth0.mc_forwarding = 0
net.ipv6.conf.eth1.forwarding = 0
net.ipv6.conf.eth1.mc_forwarding = 0
net.ipv6.conf.lo.forwarding = 0
net.ipv6.conf.lo.mc_forwarding = 0
[root@server.vabazlov.net services]# echo "net.ipv4.ip_forward = 1" > /etc/sysctl.d/90-forward.conf
[root@server.vabazlov.net services]# sysctl -p /etc/sysctl.d/90-forward.conf
net.ipv4.ip_forward = 1
[root@server.vabazlov.net services]# firewall-cmd --zone=public --add-masquerade --permanent
success
[root@server.vabazlov.net services]# firewall-cmd --reload
success
[root@server.vabazlov.net services]#
```



```
[vabazlov@client.vabazlov.net ~]$ ping www.yandex.ru
PING www.yandex.ru (77.88.55.88) 56(84) bytes of data.
64 bytes from yandex.ru (77.88.55.88): icmp_seq=1 ttl=254 time=9.39 ms
64 bytes from yandex.ru (77.88.55.88): icmp_seq=2 ttl=254 time=9.85 ms
64 bytes from yandex.ru (77.88.55.88): icmp_seq=3 ttl=254 time=11.0 ms
64 bytes from yandex.ru (77.88.55.88): icmp_seq=4 ttl=254 time=10.3 ms
64 bytes from yandex.ru (77.88.55.88): icmp_seq=5 ttl=254 time=10.5 ms
64 bytes from yandex.ru (77.88.55.88): icmp_seq=6 ttl=254 time=10.4 ms
^C
--- www.yandex.ru ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 9034ms
rtt min/avg/max/mdev = 9.391/10.226/10.954/0.496 ms
[vabazlov@client.vabazlov.net ~]$
```

Рис. 5: Доступ в Интернет

Итоги работы

- Создана пользовательская служба Firewallld
- Настроено перенаправление порта
- Включён forwarding и masquerading
- Настройки интегрированы в Vagrant provisioning
- Подтверждена корректная работа сети и механизмов Firewallld